# TEPA: Tokenized Economic Participation Agreement

White paper, draft for review · Oct 13, 2026 · @Anjal Magar

*Draft for review. Not legal, tax, or investment advice. Not an offer of securities. Review record: legal ☐ accounting ☐ technical ☐ editorial ☐*

> Prototype only. Fictional issuer and investors. No real securities are offered, no real money is accepted, and testnet tokens have no financial value.

*A note on evidence.* This paper keeps two kinds of number apart. Figures from the fictional example (the issuer, its financials, its holders) are invented to exercise the mechanism and say nothing about real demand or returns. Results from dated test runs and chain records are stated as recorded, with their date. Where a design is only proposed, or a question is still open, the text says so.

*A note on authorship.* This paper was drafted with generative AI from a source brief prepared by the project team; see "Use of generative AI" at the end.

## Abstract

The Tokenized Economic Participation Agreement (TEPA) is a proposed contract through which an established small business could raise capital by giving holders a defined percentage of its distributable cash and of its net proceeds if it is sold. Holders receive no shares, votes, board seats, or operating control. The signed agreement creates the holder's rights. Restricted, issuer-specific participation units record each holding on a blockchain and handle the administration: record-date snapshots, funded distributions, optional buyback windows, and settlement on acquisition. Units cannot be moved from one holder to another.

We describe the instrument, its economics, and the legal design it is built around, and we report on a working prototype. The prototype runs the full lifecycle for one fictional US software company and two fictional holders: reconciled sample financials, simulated eligibility checks, issuance of 600 and 400 units, a funded quarterly distribution, an oversubscribed buyback that pays and burns units in one transaction, and an acquisition settlement that ends with zero units outstanding. On 12 October 2026 a scripted rehearsal of this lifecycle passed 47 of 47 checks on a local chain with mock test USDC. The restricted token and its issuance are also deployed and source-verified on the Base Sepolia public testnet.

The live product is designed toward a US Regulation Crowdfunding offering through a registered intermediary. That route, the instrument's legal classification, and every binding commercial term remain subject to review by securities counsel.

# 1. Introduction

Consider the founder of a profitable, recurring-revenue software business who wants growth capital. The usual options fit poorly. Selling equity gives up ownership, votes, and often board influence, and it needs a valuation and an exit story that a small, durable business may not have. Debt or revenue-based financing imposes fixed repayments or fixed caps, whether the business has a good quarter or a bad one.

The investor in a small private company has the opposite problem. A rising valuation is a gain on paper only. The company may never list or be sold, and there is usually no way to receive cash or to exit. The investor also often lacks a clear record of what they are owed and why.

TEPA tests whether three mechanisms, used together, can offer useful economic participation without transferring shares or imposing scheduled debt payments:

1. variable, periodic cash distributions from a *defined* cash base;
2. optional, budget-limited buyback windows funded by the issuer;
3. contractual settlement if the company is acquired.

Tokenization is not the product. It is the record-keeping and servicing layer that makes each allocation traceable. In the words of the project plan, it "helps maintain a traceable unit registry and calculate allocations; it does not itself establish the value or enforceability of the claim."

The intended issuers are profitable, founder-led software companies with roughly $20,000 to $200,000 in monthly recurring revenue, raising from angel investors and customers who already know them. This is a proposal. Demand has not yet been tested with real founders or investors.

The rest of the paper proceeds as follows. Section 2 defines the instrument. Section 3 follows one fictional holder through the full lifecycle, introducing each formula where it is used. Section 4 explains why the units are restricted tokens. Section 5 reports what the prototype does today, and Section 6 describes how it is built. Section 7 sets out the legal design, Section 8 the risks and open questions, and Section 9 the roadmap.

# 2. The instrument

## 2.1 Definition

TEPA is a proposed contractual way for a business to raise capital in exchange for a defined percentage of its distributable cash and net acquisition proceeds, without transferring legal shares or voting control to holders.

Four points follow from that definition. The agreement defines the holder's economic claim and its limits. Restricted participation units record allocations and support administration of that claim. Distributions depend on defined cash, eligible holdings, approvals, and funding, and they can be zero. The unit is not a share, not a promise to repay a loan, and not a freely tradable asset.

The aim is to make a performance-linked, non-voting claim on a small business *legible*, *verifiable*, and *administrable*. Legible means the holder can see the formula, its inputs, its exclusions, and the risks before committing. Verifiable means every amount paid traces back to source financial records, a versioned formula, named approvals, and a chain receipt. Administrable means the issuer can run snapshots, distributions, buybacks, and a final settlement without a spreadsheet register drifting away from reality.

## 2.2 What a holder gets, and what they do not

| A participation unit gives | A participation unit does not give |
|---|---|
| A portion of defined distributable cash, when available and funded | Legal shares, votes, or board control |
| A portion of defined net acquisition proceeds under the agreement | Guaranteed payments or protection of the amount paid |
| The chance to take part in a buyback, only if the issuer offers one under approved terms | Redemption on demand, or free transfer to another investor |

In economic terms the instrument resembles equity in some respects, because what a holder receives depends on business performance and sale value. It is not legal equity. Nor is it a conventional loan: the current design has no repayment schedule, no fixed interest, and no unconditional repayment at maturity.

## 2.3 The fictional example

The rest of the paper uses one fictional issuer, Example SaaS, Inc., with 12 months of synthetic history. It issues 1,000 whole participation units, in one class, at an initial unit price of $100, for an illustrative raise of $100,000. Together the units carry 10% of defined

distributable cash and of defined net acquisition proceeds. Two fictional holders take part: Holder A with 600 units and Holder B with 400. Secondary trading is disabled.

### 2.4 Three values kept apart

A holder can meet three different numbers that each look like "the price of a unit". The prototype keeps them apart on every screen, because confusing them is how a holder comes to mistake an estimate for money they can withdraw.

| Value | What it is | What it is not |
|---|---|---|
| Initial unit price ($100) | The price paid at original issuance | A current market price |
| Informational valuation estimate | The output of an independent appraisal; non-binding | Cash, a balance, or a right to redeem |
| Binding buyback price | The price fixed for one approved, funded buyback window | A standing offer, or a promise of future windows |

The unit price is not derived mechanically from annual recurring revenue. Doubling revenue does not double the value of a unit.

## 3. One holder, from issuance to settlement

The clearest way to see how TEPA works is to follow one holder through every stage. We follow Holder A in the fictional example. All amounts are invented to exercise the mechanism; they are not a forecast, and they should not be added up into a "return".

All arithmetic in the prototype is exact. Money is held as whole minor units (cents, or six-decimal units for test USDC), percentages as basis points, and no floating-point arithmetic touches money.

### 3.1 Issuance

Holder A acknowledges the prototype terms and six disclosures, passes a simulated eligibility check, binds a wallet with a signature, and receives 600 units at the $100 initial price. Holder B receives the other 400. The chain balances and the total supply of 1,000 are then reconciled against the off-chain holder register.

### 3.2 A quarterly distribution

Distributions are paid from *defined distributable cash*, not from accounting profit, EBITDA, revenue, or a bank balance. The proposed formula is:

```
  Eligible cash receipts
- Approved operating expenses
- Taxes paid or reserved
- Required debt service
- Operating reserve contribution
- Excluded or extraordinary items
= Defined distributable cash
```

```
Holder pool = max(0, defined distributable cash) × participation rate
```

Each line item carries a category, source, period, evidence reference, and reconciliation status. The system flags related-party payments, changes in founder pay, unusual expenses, revenue reversals, loans, and manual adjustments for human review. The final legal definition belongs to counsel and accountants.

For the first quarter of 2025, the fictional issuer's figures are:

| Item, 2025-Q1 | Amount |
|---|---|
| Eligible receipts (Jan to Mar) | $31,500.00 |
| Operating expenses | −$15,300.00 |
| Taxes | −$1,800.00 |
| Debt service | −$1,200.00 |
| Reserve contribution | −$3,000.00 |
| **Defined distributable cash** | **$10,200.00** |
| **Holder pool at 10%** | **$1,020.00** |

The pool is divided by each holder's units at a fixed record block:

```
Entitlement(holder) = pool × holder's eligible snapshot units / total
eligible snapshot units
```

Each entitlement is rounded down to the minor unit, and any leftover cents go to the largest remainders, with holder ID as the tie-breaker. Entitlements always add up exactly to the approved pool. Holder A's 600 of 1,000 units give $612.00, paid as 612 mock test USDC; Holder B receives $408.00. Holder A claims once. A second claim fails on chain.

### 3.3 An attempted transfer

Holder A then tries to send units to Holder B. Both wallets are eligible, but the transfer reverts at contract level with `RestrictedTransfer`. Section 4 explains why.

### 3.4 A buyback window

Later the issuer obtains an independent appraisal and opens a buyback window. The appraisal records its date, method, inputs, and adjustments, and states whether TEPA claims are already deducted so that nothing is counted twice.

```
Indicative unit value = adjusted equity value × participation rate /
outstanding eligible units
```

```
Capacity = floor(buyback budget / binding unit price)
```

If holders tender more than the capacity, each holder's accepted units are scaled down in proportion, rounded down, with leftovers assigned by largest remainder.

| Item | Value |
|---|---|
| Appraised adjusted equity value, before TEPA | $2,000,000 |
| Indicative unit value (10% ÷ 1,000 units) | $200, informational only |
| Binding buyback price for this window | $200 |
| Issuer's buyback budget | $50,000 |
| Capacity | 250 units |
| Tendered | A: 200, B: 200 (400 in total) |
| Accepted | A: 125, B: 125 |
| Paid | $25,000 each |
| Units remaining after burn | A: 475, B: 275 (750 in total) |

The point is that the budget can be smaller than demand. Holders asked to sell 400 units and the window took 250. Payment and burning of the accepted units happen in the same transaction.

### 3.5 Acquisition settlement

Finally, the fictional company is sold.

```
  Gross sale consideration
- Transaction costs
- Debt and senior obligations
- Taxes or required reserves
- Other specifically disclosed deductions
= Net acquisition proceeds
```

```
Settlement pool      = max(0, net acquisition proceeds) × participation rate
Settlement(holder)   = settlement pool × holder's remaining units / total
remaining units
```

| Item | Amount |
|---|---|
| Gross cash consideration | $1,000,000 |
| Transaction costs | −$50,000 |
| Debt and senior obligations | −$100,000 |
| Taxes and required reserves | −$50,000 |
| **Net acquisition proceeds** | **$800,000** |
| **TEPA settlement pool at 10%** | **$80,000** |
| Holder A, 475 of 750 units | $50,666.67 |
| Holder B, 275 of 750 units | $29,333.33 |

The rounding is exact. Holder A's raw share is 5,066,666.67 cents, rounded down to 5,066,666, plus the one leftover cent by largest remainder, giving 5,066,667. Holder B receives 2,933,333. The two sum to 8,000,000 cents. Each payout burns the matching units, supply reaches zero, and the token enters a terminal state in which no new minting or distribution is possible.

A careful reader will notice that the appraisal valued the company at $2,000,000, yet it sold for $1,000,000 gross. Holders who sold into the buyback received $200 per unit; those who held to the end received about $106.67. This is deliberate in the example. An informational estimate is not a promise of what a sale will bring, and this is exactly why the paper keeps the three values in Section 2.4 apart.

Earn-outs, escrows, holdbacks, non-cash consideration, founder rollover, and post-closing adjustments are open questions. The prototype does not invent a treatment for them.

### 3.6 Holder A at a glance

| Stage | Holder A |
|---|---|
| Issuance | 600 units at the $100 initial price |
| Q1 distribution | $612 claimed once; a duplicate claim fails |
| Transfer to Holder B | Reverts with `RestrictedTransfer` |
| Buyback | 200 tendered, 125 accepted, $25,000 paid, 125 units burned |
| Acquisition | 475 remaining units settle for $50,666.67 and are burned |
| End | 0 units; token in terminal state |

## 4. Why a restricted token

Every payment in Section 3 depends on knowing exactly who held what at a record date. A spreadsheet register can drift; a restricted token gives the issuer a balance record that cannot. Units cannot be traded or moved between holders, and only the issuer-side registry can mint, recover, or burn them.

### 4.1 What the chain adds

- A tamper-evident, timestamped balance record that the off-chain register is reconciled against.
- Record-date snapshots fixed at a specific block, so the denominator of a distribution can be proved.
- Entitlements committed as a Merkle root, with one claim per entitlement and on-chain rejection of duplicate or borrowed proofs.
- Delivery against payment: a buyback or acquisition payout and the matching burn happen in one transaction.
- A terminal state after acquisition that blocks new minting on chain, independently of the application.

The token does not create rights, verify financial records, establish value, provide liquidity, or make the agreement enforceable. Those remain with the agreement, the issuer's reporting, and the law.

## 4.2 Why restricted

Three reasons point the same way. Regulation Crowdfunding imposes a one-year resale restriction. Lawful transfer exceptions need human review. And free transferability would raise broker, exchange, and alternative-trading-system questions that TEPA is designed to avoid. Blocking ordinary transfers at contract level, even between eligible wallets, keeps the token inside the agreement's boundaries.

| From | To | Allowed | Condition |
|---|---|---|---|
| Mint | Approved holder | Yes | Authorized issuance within the 1,000-unit cap |
| Holder A | Holder B | **No** | Always rejected, even if both are eligible |
| Holder | Buyback vault | Yes | Approved tender settlement only, burned on payment |
| Holder's old wallet | Holder's new wallet | Authorized only | Approved recovery case |
| Holder | Acquisition vault | Yes | Approved final settlement only |
| Any | Burn | Authorized only | Buyback, acquisition, or correction |

In the token, `transfer`, `transferFrom`, and `approve` revert unconditionally with `RestrictedTransfer()`, `allowance` returns zero, and `canTransfer` returns false.

## 4.3 Relationship to ERC-3643

ERC-3643 is a finalized Ethereum standard for permissioned tokens, with interfaces for identity, eligibility, transfer control, freezing, and recovery. TEPA's token is a custom, project-owned contract in the style of ERC-3643. It implements an eligibility check, pause, address and partial freezes, case-bound recovery, and a restricted ERC-20 read surface. It does not implement the full ONCHAINID identity suite or batch operations, and it does not claim conformance with the standard. No third-party ERC-3643 code was copied.

ERC-1400 was considered and set aside because its partitions serve multiple share classes and TEPA has one. Solana's Token-2022 can support permissioned tokens but would need a separate, Solana-specific design.

# 5. What works today

The whole lifecycle in Section 3 runs end to end. On 12 October 2026, after a fresh reset and local deployment, the strict rehearsal script passed 47 of 47 checks on a local Anvil chain (chain ID 31337) with a disposable PostgreSQL database, in about 95 seconds. It was rerun after the final restructuring of the guided demo, with the same result.

## 5.1 The full-flow rehearsal

| Stage | Recorded result, 12 October 2026 |
|---|---|
| Reconciliation | Approval refused while 3 review cases were open; the issuer's preparer could not resolve them; the approved annual pool was 624,000 cents |
| Issuance | An unapproved wallet was refused; 600 and 400 units issued; chain balances and the supply of 1,000 matched the register |
| Prohibited actions | `transfer`, `transferFrom`, `approve`, and a mined transfer all reverted with `RestrictedTransfer`; a mint by a holder reverted with `Unauthorized` |
| Distribution, 2025-Q1 | The 102,000-cent pool split 61,200 and 40,800; Holder B could not claim with Holder A's proof; both claims paid 612 and 408 mock USDC; duplicates reverted |
| Buyback | Self-approval refused; a 401-unit tender refused; early close refused; 125 and 125 accepted for $50,000; the price stayed "funding pending" until funded; a duplicate settlement reverted; 750 units remained |
| Acquisition | The $80,000 pool settled over 750 units as 5,066,667 and 2,933,333 cents; supply reached zero; a new distribution and a new mint were both refused |
| Audit | 14 chain-backed audit events carried both command and chain-event references; a holder was refused access to the audit explorer |

Many of these checks test what the system *refuses* to do. For a financial instrument, the refusals matter as much as the payments.

One limit applies: the script signs holder actions with local test keys through the API. It is not yet a rehearsal with a browser wallet extension.

## 5.2 Test suites

| Suite | Result, 12 October 2026 |
|---|---|
| Domain unit tests | 23 passed |
| API unit tests | 21 passed |
| API, PostgreSQL, and Anvil integration | 35 of 35, including headless-browser claim and settle clicks and checks at 375 px width |
| Web tests | 31 passed |
| Foundry contract tests | 13 of 13 across four suites |
| Format, lint, type check, fixtures, build | Passed |

These runs were local. No remote continuous-integration run is recorded.

## 5.3 Public evidence on Base Sepolia

The restricted token is deployed on the Base Sepolia public testnet (chain ID 84532) at `0xD4685cF0aeC08A86C5aB9Dc86cd177FC4775B663`. Anyone can check it.

| Evidence | Value |
|---|---|
| Source verification | Exact match; compiler v0.8.24, 200 optimizer runs, EVM cancun |
| Deployed | 24 September 2026, block 47260215 |
| Holder A mint, 600 units | Block 47293065 |
| Holder B mint, 400 units | Block 47293074 |
| Supply and decimals | 1,000 of 1,000 maximum; 0 decimals |
| Transfer restriction | `transfer`, `transferFrom`, and `approve` revert with `RestrictedTransfer()` |
| Reconciliation, rechecked 12 October 2026 | Register and chain balances matched; zero mismatches |

This evidence covers **issuance only**. No distribution, buyback, or acquisition has taken place on Base Sepolia; those flows ran on the local chain with mock test USDC.

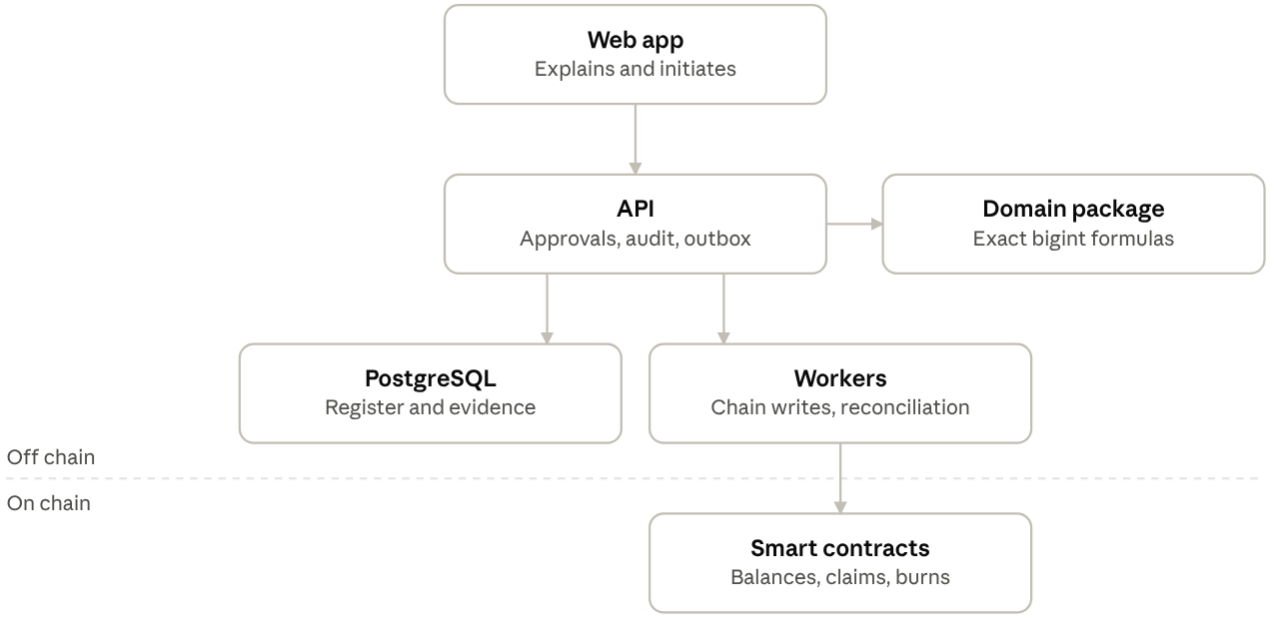
### 5.4 Known limitations

- The contracts are prototypes and have not been audited.
- One local test key holds every contract role. On Base Sepolia, one wallet holds the admin, compliance, issuer-agent, pauser, and recovery roles.
- Acquisition finalization needs every remaining holder to settle; a holder who never settles blocks closure.
- Unclaimed or blocked funds cannot be swept; the policy for them is an open legal question.
- Eligibility, identity checks, sanctions screening, subscriptions, approvals, and the appraisal are simulated.
- The Base Sepolia deployment predates the repository's Git history, so it records a source fingerprint rather than a commit.
- A real wallet-extension rehearsal, keyboard-only navigation, and screen-reader checks are still to do.

## 6. System design

The prototype separates what explains, what decides, what calculates, and what records. Each part is the source of truth for one thing only.

**Formulas live in one package; balances live on chain**




TEPA prototype components · 6 parts, off chain and on chain

The web app (React, TypeScript) explains and initiates but holds no formulas. The API (Node.js, Express) enforces roles, workflow states, approvals, audit events, and a durable outbox. The domain package computes every formula with exact `bigint` arithmetic and versioned rounding rules. PostgreSQL holds the off-chain register and its evidence. Workers write to the chain, confirm receipts, and reconcile the two records. The Solidity contracts (0.8.24, tested with Foundry) hold restricted balances, claims, and settlement receipts.

## 6.1 The contracts

| Contract | Lines | Role |
|---|---|---|
| `TepaToken.sol` | 300 | Restricted whole-unit token with a 1,000-unit cap; eligibility, mint, pause, freeze, recovery, settlement burn, terminal marker |
| `DistributionVault.sol` | 124 | One immutable vault per quarter; fixed Merkle root; funded before claims open; one claim per entitlement |
| `SettlementVault.sol` | 173 | One immutable vault per buyback or acquisition; pays test USDC and burns units in one transaction |
| `TestUsdc.sol` | 46 | Local-only, freely mintable six-decimal mock; never a hosted asset |

The token separates eight scoped roles, including pauser, compliance, recovery, distribution, buyback, and settlement. In a production design no single account would hold them all. In the demo they are concentrated in one key, and Section 5.4 says so.

## 6.2 What goes on chain and what stays off

| On chain | Off chain |
|---|---|
| Wallet eligibility state (a pseudonymous reference, expiry, jurisdiction code) | Agreement versions and acknowledgements |
| Restricted balances | Holder register and identity references |
| Entitlement Merkle roots and claim state | Financial inputs, reconciliation, approvals |
| Settlement payments and burns | Calculations with formula versions and rounding traces |
| Lifecycle events with case IDs | Append-only audit log |

No names, identity documents, bank records, customer records, or detailed financials go on chain.

## 6.3 Design choices

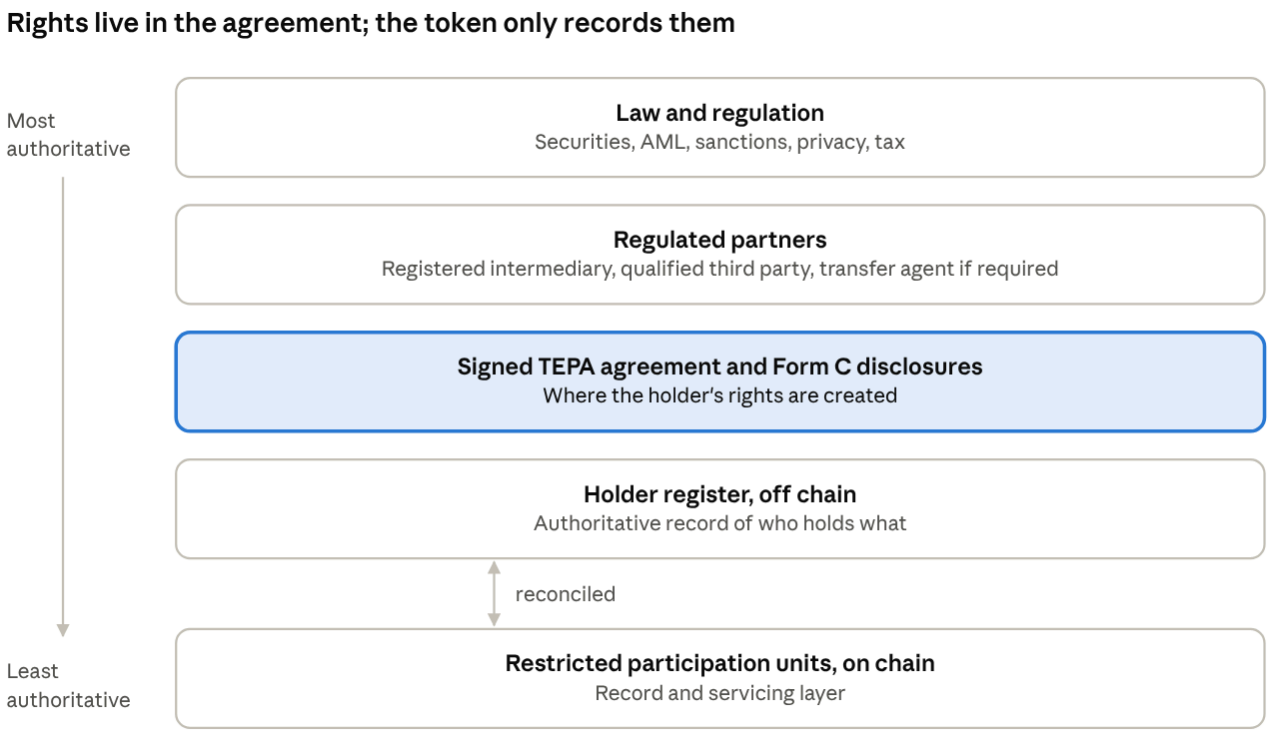
- **Immutable contracts, no proxies.** A change in behaviour means a new deployment, so an upgrade cannot alter entitlements already earned.

- **One vault per period or settlement case.** Its Merkle root cannot be replaced, and claims cannot open while it is underfunded.
- **No sweep function.** Unclaimed or blocked funds stay committed until a reviewed policy exists.
- **A hash is not a settlement.** A payment counts as settled only once the system has matched the holder, period or case, leaf index, amount, vault event, block hash, and log index.
- **Mismatches are surfaced, never repaired silently.** Any failed cross-system write, or any disagreement between the register and the chain, becomes `RECONCILIATION_REQUIRED` for a person to resolve.
- **No self-approval.** Preparer, finance reviewer, compliance operator, valuation reviewer, distribution operator, registry agent, holder, and auditor are separate roles. A changed financial input invalidates earlier approvals, and the old calculation stays visible.

The contracts also use custom errors, checks-effects-interactions ordering, reentrancy protection, and safe transfers, and they avoid `tx.origin` and unbounded loops over holders.

## 7. Legal design

TEPA is built agreement-first. The signed agreement carries the holder's rights; the token records the holding and services the claim. Everything else in the design follows from that order.




TEPA legal stack · most to least authoritative

Three consequences follow. First, the token's code, metadata, or interface never defines a right; unit metadata refers to the hash of the approved agreement and disclosure. Second, if the agreement is silent on an edge case, the software does not invent a formula; the project rule is that "code must not fill silence in the agreement". Third, software implements controls derived from law but is not itself bound by a statute. A passing test proves a property of the implementation, not a legal conclusion.

## 7.1 Working assumption: treat TEPA as a security

For design and risk management, the project treats TEPA as a security. Tokenization does not change how securities law applies, and calling something "non-equity" or a "token" does not take it outside that law. Depending on its final terms, the instrument might be classed as a contractual participation security, an investment contract, a profit-participation interest, or something else. One question is flagged for counsel in particular: whether a return linked to company value or sale events could be treated as a security-based swap.

## 7.2 Proposed route: US Regulation Crowdfunding

The live product is designed toward an offering under Regulation Crowdfunding (Reg CF, 17 CFR Part 227), subject to review of this exact instrument by US securities counsel and a prospective registered intermediary. This is a design direction, not a claim that the

instrument qualifies. If the final terms do not fit Reg CF, the instrument and route will be reassessed before any live investment flow is built.

Reg CF would shape the product in specific ways. Every live offer and commitment would run through one SEC-registered broker-dealer or funding portal; there would be no direct checkout with the issuer. Investor funds would go to the intermediary's qualified third party, not to the issuer's wallet or to a smart contract. The issuer would file Form C disclosures and annual reports. Investment limits for non-accredited investors would be decided or confirmed by the intermediary, not inferred from a wallet balance. The one-year resale restriction is why transfers are blocked by default. The offering cap, currently $5 million in 12 months [to verify against the current rule before publication], would be a versioned setting, not a constant in code.

## 7.3 What stays outside TEPA

TEPA does not match buyers and sellers, recommend transactions, take transaction-based fees, hold customer securities, or run a secondary market. It has no exchange, order book, liquidity pool, bridge, staking, or lending. Each of those would need new legal analysis and licensed partners. An ERC-3643 allowlist is not a broker or exchange licence.

Identity checks, anti-money-laundering programmes, and sanctions screening belong to the regulated partners, under a signed allocation of responsibilities. Identity documents, tax IDs, bank data, and screening results stay encrypted off chain; only pseudonymous references and integrity hashes go on chain. The prototype never says that a holder "passed KYC"; it records a mock decision reference, an expiry, and a jurisdiction code.

Whether the chain, an off-chain register, or a combination is the legal record of ownership, and whether a registered transfer agent is required, are open decisions. Until they are settled in signed documents, the project will not describe the blockchain as the legal register. Wallet control is not legal ownership.

## 7.4 From law to code

The repository maps each legal requirement to a stable control ID, an owner, a code control, a negative test, and evidence. There are 49 controls across seven families: core securities law (7), Reg CF issuer duties (14), intermediary-owned duties (7), token and registry (6), contract terms to resolve (6), identity, AML, sanctions and funds (5), and a conditional UK perimeter (4). Every control is marked design-only for the prototype.

The controls are enforced as state guards in code. Two examples:

```
can_mint = partner_closing_confirmed && funds_flow_confirmed &&
allocation_final
  && cancellation_reconfirmation_complete && registry_case_approved
  && chain_registry_reconciled

can_distribute = signed_formula_version_approved && reconciled_inputs
  && dual_review && snapshot_immutable && lawful_funding_route_confirmed
  && payout_fully_funded
```

These guards are necessary engineering gates, not sufficient legal conditions. The prototype substitutes explicit demo events and simulated partner decisions for them, and its startup configuration refuses any flag for real money, real providers, or secondary transfers. There is deliberately no simple "production enabled" switch. Going live would require a signed release artifact with named approvals, contract addresses, the Git commit, and the terms and disclosure versions.

# 8. Risks and open questions

## 8.1 Risks to holders

A holder could lose the entire amount paid. Distributions can be zero in any period, and nothing is guaranteed. Units are illiquid: there is no secondary market, transfers are blocked, and buybacks are optional and limited by budget. Holders depend on the issuer's honesty, reporting, and solvency, and they have no control or votes. An informational valuation is not cash. The effects of dilution or reorganization, the holder's information rights, and tax treatment are all still unresolved.

## 8.2 Issuer conduct and how the design responds

| Risk | How the prototype responds |
|---|---|
| Revenue is unverified or omitted | Processor, bank, and ledger records are compared, and mismatches are flagged for review |
| Expenses are inflated to shrink distributions | A fixed, disclosed cash definition; flags on related-party and unusual expenses; approvals invalidated when inputs change |
| Valuation is inflated | An independent appraisal step, separate from reported revenue, with its assumptions shown |

| Risk | How the prototype responds |
|---|---|
| Holders mistake estimates for liquidity | Three values kept apart; a buyback budget visibly smaller than demand |
| No reliable record of rights | Wallet, balance, subscription, and off-chain register linked and reconciled |
| Admin powers are misconfigured | Tests for direct transfers and revoked wallets; role changes in the audit log |
| Wrong snapshot or double claim | A fixed record block; Merkle-committed entitlements; one claim per entitlement |
| The instrument cannot be offered as drafted | Counsel's classification and a registered intermediary are launch gates |

## 8.3 Technical and regulatory risks

On the technical side: defects in unaudited contracts, compromise of admin keys, concentration of roles, lost holder keys, chain reorganizations, RPC outages, sequencer downtime on a layer-2 chain, a stablecoin freeze or loss of peg, and mismatches between the database and the chain. On the regulatory side: the instrument could be classified differently, may not fit Reg CF, may raise security-based swap questions, may require a transfer agent, and is exposed to state-law issues and to changes in law or SEC guidance.

## 8.4 Decisions not yet made

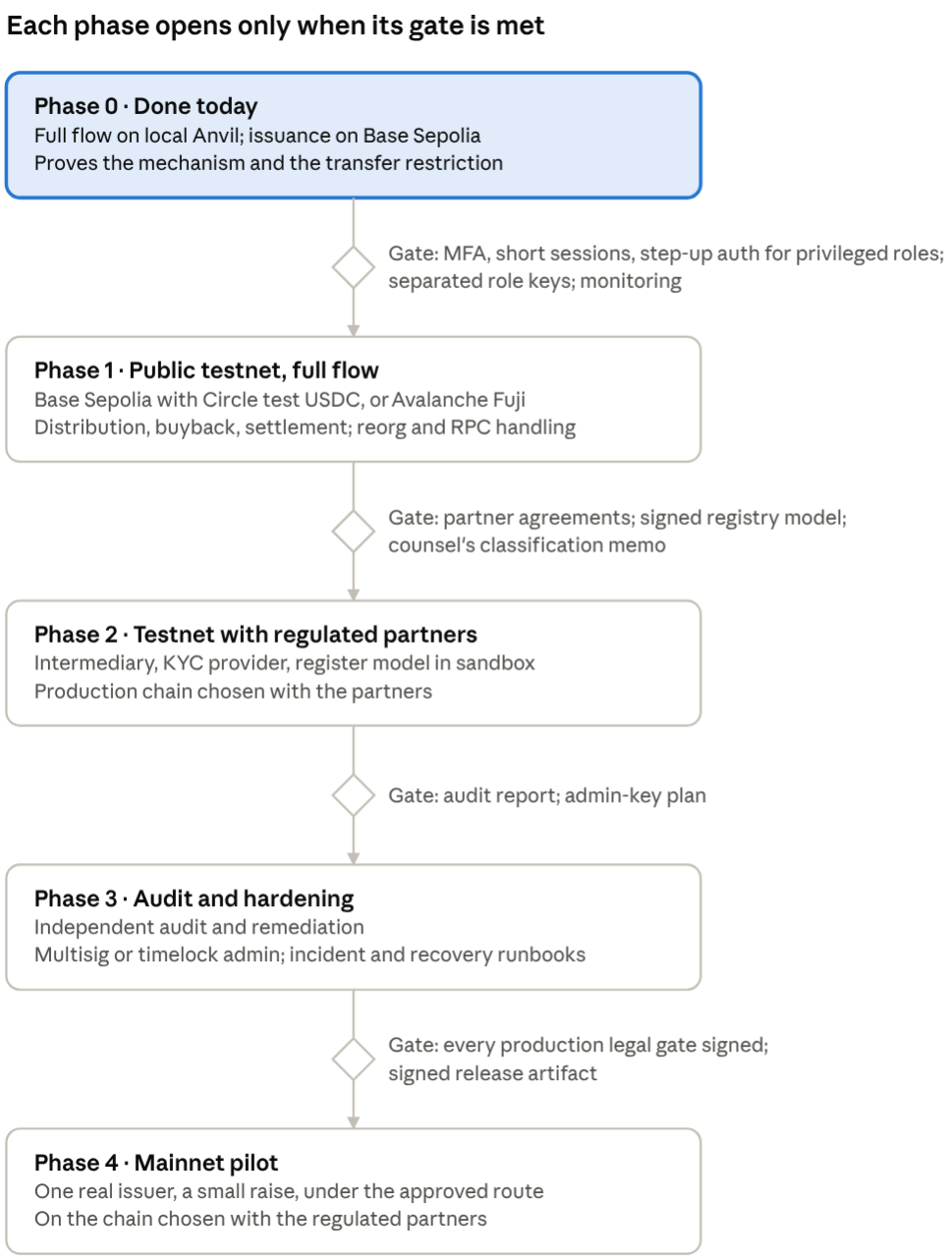
The following are deliberately left open. The prototype does not answer them, and no legal decision has yet been recorded.

1. Final legal classification and name of the instrument.
2. Issuer's state of incorporation and governing law.
3. The precise definition of distributable cash and the reserve policy.
4. Seniority, subordination, insolvency treatment, and remedies.
5. Whether participation covers company-wide cash, one product line, or another base.
6. Treatment of new debt, new equity, further TEPA issues, reorganizations, and dilution.
7. Whether past distributions count in later valuation or acquisition calculations.
8. The acquisition waterfall for earn-outs, escrows, rollover, non-cash consideration, and partial sales.
9. Mandatory or optional buybacks.
10. The authoritative holder record and the transfer-agent model.
11. Custodial or self-custodial holder wallets.

12. The process for lawful transfer exceptions after the restriction period.
13. Stablecoin or fiat payouts, and the custody model.
14. Unclaimed, blocked, rejected, or sanctioned-holder funds.
15. Tax classification, withholding, and reporting.
16. Amendment thresholds and holder consent rights.
17. Disaster recovery if the chain or contracts are unavailable.
18. Jurisdictions and investor categories allowed at launch.
19. The production chain (see Appendix C).

## 9. Roadmap

Mainnet is a possible future step, not a current one. Each phase opens only when the gate before it has been met, and no dates are promised.




TEPA roadmap · 5 phases, 4 gates, no dates

Each chain gets its own deployment record, keyed by chain ID and commit. A testnet address is never reused on mainnet, and every deployment checks the chain ID, bytecode, and the settlement asset's address, symbol, and decimals before it broadcasts.

Alongside the phases, four streams of work continue. **Prototype:** a rehearsal with a real browser wallet, keyboard and screen-reader checks, and publishing the repository after a review for secrets. **Validation:** interviews with founders and angel investors to test whether a 10% cash claim is attractive, whether buyback windows and acquisition settlement answer their liquidity concerns, and what reporting holders need. None has been done yet. **Partners:** securities counsel, a registered intermediary, a register or

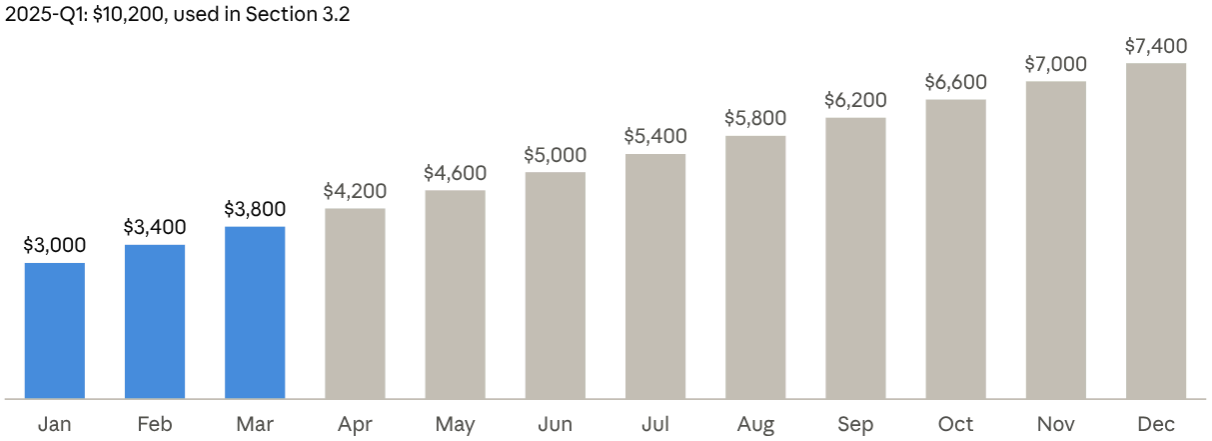
transfer-agent model, an identity and AML provider, payment rails, and accounting integrations. **Hardening:** an independent audit, multisig or timelock administration, separated role keys, monitoring, and incident, recovery, and wind-down runbooks.

## Appendix A. The fictional year in full

The fictional issuer's defined distributable cash for the 12 months of 2025 totals $62,400, giving a holder pool of $6,240 at 10%.

**Fictional defined cash for 2025; the first quarter funds the example distribution**

Fictional synthetic data, not a forecast · $62,400 for the year · holder pool at 10%: $6,240




Fictional fixture data, formula version sample-distributable-cash-v1 · 12 months of 2025

| Line, 12 months of 2025 (fictional) | Amount |
|---|---|
| Eligible receipts | $153,000.00 |
| Operating expenses | −$66,600.00 |
| Taxes | −$7,200.00 |
| Debt service | −$4,800.00 |
| Reserve contribution | −$12,000.00 |
| Excluded items | $0.00 |
| **Defined distributable cash** | **$62,400.00** |

In every buyback, accepted units never exceed what a holder tendered, what they hold, the window's capacity, or the funded budget. Where a distribution has no eligible units, the result is defined explicitly rather than calculated.

# Appendix B. Production legal gates

No real offer, sale, mint, payment, distribution, buyback, acquisition settlement, or public production deployment may happen until each gate below has written approval. None has been met.

| Gate | Evidence required |
|---|---|
| Instrument classification | Written analysis by US securities counsel, including the security-based swap question |
| Offering exemption | Reg CF eligibility and offering plan approved by counsel and the intermediary |
| Intermediary | An executed agreement with an SEC-registered broker-dealer or funding portal |
| Funds flow | An approved closing and refund process through a qualified third party |
| Ownership record | A signed decision on the master holder record and transfer-agent duties |
| Identity, AML, sanctions | A responsibility matrix, vendor contracts, policies, escalation, and testing |
| Corporate authority | Board or member approvals and executed instrument documents |
| Disclosure | An approved Form C package, risk factors, communications, and promoter process |
| Accounting and tax | Qualified review of the instrument, distributions, withholding, and reporting |
| Privacy | Data map, notices, contracts, retention schedule, and risk assessment |
| Smart contracts | Independent audit, remediation, deployment controls, and an admin-key plan |
| Operations | Runbooks for reconciliation, incidents, recovery, complaints, disputes, and wind-down |
| Jurisdictions | Approved investor and issuer geographies; non-US access addressed |

UK rules would become relevant only if a UK entity ran the platform, UK investors were approached, UK promotions were accessible, or UK personal data were processed. A US Reg CF offering does not authorize a UK offering.

## Appendix C. Choosing a production chain

TEPA's on-chain workload is small: one issuance batch, about four distributions a year, an occasional buyback window, and one final settlement. Throughput and fees are therefore not the deciding factors. What matters is finality and reorg risk, native USDC on the chain, which chains the regulated partners support, who orders transactions, privacy, EVM compatibility, and tooling for verification and audit.

*The chain facts below come from general knowledge as of mid-2026 and have not yet been checked against primary sources. They must be verified and dated before publication.*

| Criterion | Ethereum mainnet | Base mainnet | Avalanche C-Chain | Avalanche permissioned L1 |
|---|---|---|---|---|
| Chain ID | 1 | 8453 | 43114 | Chosen per chain |
| Testnet | Sepolia | Base Sepolia (used by TEPA) | Fuji | Fuji-based test L1 |
| Type | Proof-of-stake L1 | Optimistic rollup on Ethereum; single sequencer | EVM L1 | Own EVM chain with its own validators |
| Typical finality | Minutes | Fast soft confirmation; L1-backed later; 7-day withdrawal window | About 1 to 2 seconds | Depends on validator set |
| Native USDC | Yes | Yes | Yes | Not by default |
| Operator trust | Most decentralized | One sequencer, with L1 forced inclusion | Public validator set | Highest; chosen validators |
| Effort to move from today | Low (redeploy) | Lowest (same family as current testnet) | Low (redeploy) | Highest (run a chain) |

Base is the path of continuity: the token is already deployed and verified on Base Sepolia, the tooling is identical, and native USDC exists; the main risk to disclose is reliance on one sequencer. Ethereum offers the strongest settlement assurance at the highest cost,

which may be acceptable for a handful of transactions a year. Avalanche C-Chain offers fast finality, which simplifies snapshots and receipt confirmation. A permissioned Avalanche chain fits only if a partner needs control over who can transact. Because the rights live in the agreement and the register, and the contracts are plain Solidity, TEPA can move between EVM chains. The production chain should be chosen with the regulated partners, not ahead of them.

## Appendix D. Glossary

| Term | Meaning in this paper |
|---|---|
| TEPA | Tokenized Economic Participation Agreement: the signed contract that defines the holder's economic claim |
| Participation unit | One of a fixed number of whole, restricted, issuer-specific units recording a holder's portion of the claim |
| Holder | A party to the agreement who holds participation units |
| Defined distributable cash | Eligible receipts minus the agreed deductions |
| Net acquisition proceeds | Gross sale consideration minus the agreed deductions |
| Participation rate | The aggregate portion for all units; 10% in the example |
| Distribution | Payment of an approved, funded pool to eligible holders for one period |
| Record date or snapshot | The chain block at which eligible balances are fixed |
| Initial unit price | The price at original issuance |
| Informational valuation estimate | A non-binding, appraisal-based estimate; not cash or a redemption right |
| Issuer-funded buyback window | An optional, budget-limited offer by the issuer to buy units at a binding price |
| Acquisition settlement | Final payment of the TEPA portion of net proceeds, after which units are burned |

| Term | Meaning in this paper |
|---|---|
| Restricted transfer | A contract-level block on ordinary transfers between holders |
| Holder register | The authoritative off-chain record of holders, reconciled with chain balances |
| Test USDC | A mock stablecoin with no value |
| Simulated eligibility | Mock identity, sanctions, and eligibility decisions used in the prototype |
| Reg CF | US Regulation Crowdfunding, 17 CFR Part 227 |
| ERC-3643 | An Ethereum standard for permissioned tokens |

## Sources

The paper draws on the project repository as it stood on 13 October 2026: the project rules and doctrine, the plan, the design guide, the legal traceability file, the architecture and testing records, the Base Sepolia deployment record, the contract engineering decisions, and the fixture data.

External sources cited in the repository, all to be re-checked before publication:

- Regulation Crowdfunding, 17 CFR Part 227
- Securities Act and Exchange Act, Title 15 of the US Code
- SEC Division of Corporation Finance, *Statement on Tokenized Securities*, 28 January 2026 (title, date, and content to be confirmed)
- FINRA Rule 3310 and Funding Portal Rule 300; 31 CFR Part 1023; 31 CFR chapter V (OFAC)
- E-SIGN Act, 15 U.S.C. §7001; FTC Act, 15 U.S.C. §45
- ERC-3643 specification
- TepaToken on Base Sepolia
- Official Ethereum, Base, and Avalanche documentation; L2BEAT's entry for Base; Circle's list of native USDC chains (to be added for Appendix C)

## Use of generative AI

This white paper was written with the help of generative AI. The project team compiled a source brief from the TEPA repository, covering the instrument, its legal design, the prototype, and its recorded test evidence. Claude, an AI model made by Anthropic, then

organized that material, drafted the prose, and drew the figures, following the team's direction on audience and structure.

Every fact, figure, and test result in this paper comes from the team's source brief; the AI was instructed not to add claims of its own. The worked examples were rechecked arithmetically against the brief. Facts about third-party chains and external legal sources were not independently verified during drafting and are marked for verification. The project team is responsible for the content.

## Version and review

Version 1.0, 13 October 2026. Status: draft for review. Not legal, tax, or investment advice. Not an offer of securities. Illustrative numbers are fictional and are kept separate from recorded prototype evidence. Legal, accounting, technical, and editorial review: none completed.